Fiscal Year:	FY 2026	**Task Last Updated:**	FY 04/26/2026
PI Name:	Rutkove, Seward M.D.		
Project Title:	Acute and Long-Term Effects of Combined Radiation and Partial Unloading on Neurological and Musculoskeletal Systems in Male and Female Rats		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	None		
PI Email:	srutkove@bidmc.harvard.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-667-8130
Organization Name:	Beth Israel Deaconess Medical Center, Inc./Harvard Medical School		
PI Address 1:	330 Brookline Ave TCC-810		
PI Address 2:			
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02215-5400	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
Start Date:	06/01/2023	**End Date:**	05/31/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
Contact Email:	harry.jones@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	There are no changes.		
COI Name (Institution):	Bouxsein, Mary Ph.D. (Beth Israel Deaconess Medical Center, Inc.) Mortreux, Marie Ph.D. (University of Rhode Island) Lemere, Cynthia Ph.D. (Brigham And Women's Hospital, Inc.)		
Grant/Contract No.:	80NSSC23K0774		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In this proposed work, we will evaluate the separate and combined effects of simulated space radiation and lunar gravity on the muscular, skeletal, and central nervous systems over the short- and longer-term. Moreover, we will assess whether these effects differ by biological sex. This work will be accomplished by studying male and female rats of an age (in rat years) comparable to that of the astronaut corps and exposing them simultaneously to simulated galactic cosmic rays (Galactic Cosmic Rays simulation: GCRsim) at Brookhaven National Laboratory and extended 20% partial weight-bearing to mimic lunar gravity. Animals will be assessed after completing 28 days of partial weight-bearing to assess short-term effects of these exposures and several months later to assess longer-term effects with recovery. This will include a variety of functional, molecular, and histological assessments assessing bone, muscle, and brain. This project is aligned with recommendations listed in the Decadal Survey, specifically: AH2) Preservation and reversibility of bone structure and strength; AH5) understanding underlying mechanisms regulating net skeletal muscle protein balance and protein turnover during states of unloading and recovery; and AH9) Investigate the effects of prolonged reduced gravity on task completion; B2) Conduct integrated translational research in which long-duration missions are simulated specifically for the purpose of studying the interrelationships among individual functioning and cognitive performance. Deliverables: This proposal will provide new data on the sex-specific effects of combined lunar gravity and galactic cosmic ray exposure on Central Nervous System (CNS) and musculoskeletal health to help us prepare more effectively for future extended lunar missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will provide valuable insights the impact of reduced muscle loading (as would occur here on Earth when using crutches or a walker) on longer term motor function, including muscle, skeletal, and brain health and cognition. It will also evaluate the effects of radiation exposure on muscle, skeletal, and brain health and cognition, which could have relevance to patients being treated with radiotherapy for various cancers or being inadvertently exposed to radiation (in an industrial exposure or accident). Moreover, it will evaluate the effect of these two stressors together, which is certainly not an uncommon situation in patients who are debilitated while receiving radiotherapy for neoplastic disease.
Task Progress:	In the past year we have completed our second and final campaign in rats at Brookhaven National Laboratory in summer of 2025 and subsequent follow-up studies at Beth Israel Deaconess Medical Center, Boston, MA in the fall of 2025. The campaign included extensive planning and organization in the months leading up to the campaign at Brookhaven. This included building and reconfiguring more cages for the period of partial (20%) unloading for all animals, such that whether unloaded or not, the animals were in identical environments. While our first campaign focused on studying only female animals, this second campaign focused on completing male animals, with a few additional female animals to account for some minor, unexpected reduction in animal numbers during the first campaign. We have since collected all materials and have begun analysis, including histological and molecular assays, which are now currently underway.
Bibliography Type:	Description: (Last Updated: 05/21/2026)
Abstracts for Journals and Proceedings	Dunlap K, Yadav S, Wong M. Krassel C, Malone Z, Lemere C, Bouxsein M, Mortreux M, Rutkove S. "Reduced musculoskeletal loading, not radiation, drives eeductions in skeletal muscle strength and mass in male and female rats after four weeks of simulated lunar gravity and galactic cosmic radiation exposure. " 41st Annual Meeting of the American Society for Gravitational and Space Research, Phoenix, Arizona, December 3-6, 2025. Abstracts. 41st Annual Meeting of the American Society for Gravitational and Space Research, Phoenix, Arizona, December 3-6, 2025. https://asgsr.org/wp-content/uploads/2026/02/2025-Annual-Meeting-Booklet.pdf , Dec-2025